RIGHT EXACTNESS OF THE TANGENT FUNCTOR IN ITS MODULE ARGUMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement and proof use the author's own labels item-mult-map-linear and item-tangent-right-exact; both are delivered as written.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The tangent functor construction and its canonical map N tensor_B F(B) -> TF(B, N). Same chapter, Lemma lemma-tangent-functor, invoked at proof lines 1294-1297. Supplies TF, the canonical map and the results used throughout without further mention.

Source reference. Original lemma 06Z8: sources/stacks-adequate/pinned-original-source.tex, source lines 1138–1138. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let F be a module-valued functor such that for any $B \in \mathrm{Ob}(Alg_A)$ the functor $TF(B, -)$ on B-modules transforms a short exact sequence of B-modules into a right exact sequence. Then*

1. $TF(B, N_1 \oplus N_2) = TF(B, N_1) \oplus TF(B, N_2)$,
2. *there is a second functorial B-module structure on $TF(B, N)$ defined by setting $x \cdot b = TF(B, b \cdot 1_N)(x)$ for $x \in TF(B, N)$ and $b \in B$,*
3. *the canonical map $N \otimes_B F(B) \to TF(B, N)$ of Lemma 06Z8 is B-linear also with respect to the second B-module structure,*
4. *given a finitely presented B-module N there is a canonical isomorphism $TF(B, B) \otimes_B N \to TF(B, N)$ where the tensor product uses the second B-module structure on $TF(B, B)$.*

Proof. We will use the results of Lemma 06Z8 without further mention. The maps $N_1 \to N_1 \oplus N_2$ and $N_2 \to N_1 \oplus N_2$ give a map $TF(B, N_1) \oplus TF(B, N_2) \to TF(B, N_1 \oplus N_2)$ which is injective since the maps $N_1 \oplus N_2 \to N_1$ and $N_1 \oplus N_2 \to N_2$ induce an inverse. Since TF is right exact we see that $TF(B, N_1) \to TF(B, N_1 \oplus$

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$N_2) \to TF(B, N_2) \to 0$ is exact. Hence $TF(B, N_1) \oplus TF(B, N_2) \to TF(B, N_1 \oplus N_2)$ is an isomorphism. This proves (1).

To see (2) the only thing we need to show is that $x \cdot (b_1 + b_2) = x \cdot b_1 + x \cdot b_2$. (Associativity and additivity are clear.) To see this consider

$$N \xrightarrow{(b_1, b_2)} N \oplus N \xrightarrow{+} N$$

and apply $TF(B, -)$.

Part (3) follows immediately from the fact that $N \otimes_B F(B) \to TF(B, N)$ is functorial in the pair (B, N).

Suppose N is a finitely presented B-module. Choose a presentation $B^{\oplus m} \to B^{\oplus n} \to N \to 0$. This gives an exact sequence

$$TF(B, B^{\oplus m}) \to TF(B, B^{\oplus n}) \to TF(B, N) \to 0$$

by right exactness of $TF(B, -)$. By part (1) we can write $TF(B, B^{\oplus m}) = TF(B, B)^{\oplus m}$ and $TF(B, B^{\oplus n}) = TF(B, B)^{\oplus n}$. Next, suppose that $B^{\oplus m} \to B^{\oplus n}$ is given by the matrix $T = (b_{ij})$. Then the induced map $TF(B, B)^{\oplus m} \to TF(B, B)^{\oplus n}$ is given by the matrix with entries $TF(B, b_{ij} \cdot 1_B)$. This combined with right exactness of $\otimes$ proves (4). $\square$